

# Chapter 14

## The Normal Distribution

Lecture 14 | Devore 9e: 4.3

Recall the continuous toolkit from last lecture:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx, \quad \mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx, \quad \text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \mathbb{E}[X^2] - \mu^2.$$

### 14.1 Why this distribution and not another

Take  $X_1, X_2, \dots, X_N$  independent and identically distributed — *whatever* their common distribution — and form the average

$$Z = \frac{1}{N} \sum_{i=1}^N X_i.$$

As  $N$  grows, the distribution of  $Z$  looks like the same bell-shaped curve, no matter what the  $X_i$  were. Complicated inputs, one universal output. That is the central limit theorem (Lecture 21), and it is the reason the normal distribution is everywhere: any quantity that is the accumulation of many small independent effects — measurement error, height, exam totals — comes out approximately normal.

### 14.2 Definition

**Definition 14.1** (Normal distribution).  $X \sim \text{Normal}(\mu, \sigma^2)$  if its pdf is

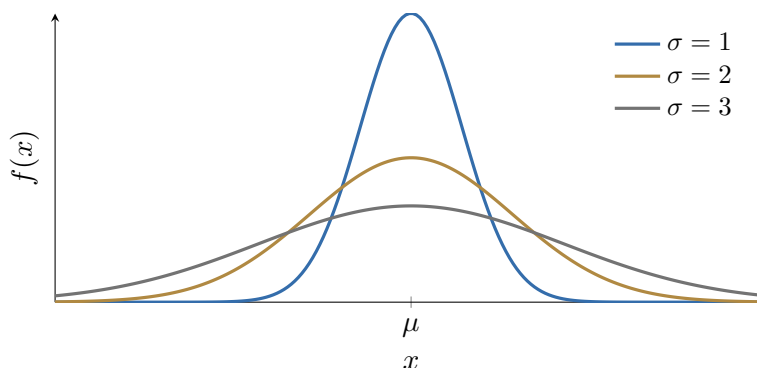
$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right), \quad x \in \mathbb{R},$$

where  $\mu \in \mathbb{R}$  and  $\sigma > 0$ .

Three facts to check (all are Homework):

$$\int_{-\infty}^{\infty} f(x; \mu, \sigma) dx = 1, \quad \mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

The factor  $\frac{1}{\sqrt{2\pi}\sigma}$  is exactly the *normalization* needed to make the total area 1; the shape is carried entirely by  $\exp(-\frac{1}{2}z^2)$  with  $z = \frac{x-\mu}{\sigma}$ .



The curve is symmetric about  $\mu$  (so the mean, median and mode all coincide there), and larger  $\sigma$  means more spread. Note that the peak height is  $\frac{1}{\sqrt{2\pi}\sigma}$ , so a very small  $\sigma$  gives a very tall narrow spike — again a reminder that a density is not a probability and may exceed 1.

### 14.3 The standard normal, and standardization

**Definition 14.2** (Standard normal).  $Z \sim \text{Normal}(0, 1)$ , with pdf and cdf

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad \Phi(z) = \mathbb{P}\{Z \leq z\} = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

**Remark 14.3.**  $\Phi$  has no closed form in elementary functions — you cannot antidifferentiate  $e^{-t^2/2}$  by hand. That is why the standard normal table exists, and why every normal computation is routed through it.

**Theorem 14.4** (Standardization). If  $X \sim \text{Normal}(\mu, \sigma^2)$  and

$$Z := \frac{X - \mu}{\sigma},$$

then  $Z \sim \text{Normal}(0, 1)$ .

*Proof.*  $F_Z(z) = \mathbb{P}\left\{\frac{X-\mu}{\sigma} \leq z\right\} = \mathbb{P}\{X \leq \mu + \sigma z\} = F_X(\mu + \sigma z)$ . Differentiate in  $z$ :  $f_Z(z) = \sigma f_X(\mu + \sigma z) = \sigma \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-z^2/2} = \varphi(z)$ .  $\square$

This single fact reduces every normal question to a table lookup:

$$\mathbb{P}\{X \leq x\} = \mathbb{P}\left\{\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right\} = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

and consequently

$$\mathbb{P}\{a \leq X \leq b\} = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

**Remark 14.5** (Reading the table backwards). Percentiles work the same way. If  $z_p$  solves  $\Phi(z_p) = p$ , then the  $p$ -th percentile of  $X$  is  $x_p = \mu + \sigma z_p$ . Values worth memorizing:

$$z_{0.90} = 1.28, \quad z_{0.95} = 1.645, \quad z_{0.975} = 1.96, \quad z_{0.99} = 2.33.$$

By symmetry  $\Phi(-z) = 1 - \Phi(z)$ , which is how negative arguments are handled.

**Example 14.6.** Exam scores are  $\text{Normal}(72, 8^2)$ . What fraction of students score between 60 and 90?

$$\mathbb{P}\{60 \leq X \leq 90\} = \Phi\left(\frac{90 - 72}{8}\right) - \Phi\left(\frac{60 - 72}{8}\right) = \Phi(2.25) - \Phi(-1.5) = 0.9878 - 0.0668 = 0.9210.$$

One more identity makes many normal expectations easy. Differentiating the density gives  $\varphi'(z) = -z\varphi(z)$ , so  $z\varphi(z)$  has antiderivative  $-\varphi(z)$  and

$$\int_a^\infty z\varphi(z) dz = \varphi(a).$$

**Example 14.7** (Only the upside). A stock trades at 100 today, and its price one month from now is modeled as  $X \sim \text{Normal}(100, 5^2)$ . A contract pays  $\max(X - 100, 0)$  dollars in one month: the gain above today's price, or nothing. What is its expected payoff?

Write  $X = 100 + 5Z$ , so the payoff is  $5\max(Z, 0)$ . By the identity with  $a = 0$ ,

$$\mathbb{E}[\max(Z, 0)] = \int_0^\infty z\varphi(z) dz = \varphi(0) = \frac{1}{\sqrt{2\pi}} \approx 0.399,$$

and the expected payoff is  $5 \times 0.399 \approx \$2.00$ , about 0.4 standard deviations. The contract is a call option, and “0.4 standard deviations” is the usual back-of-the-envelope price for one struck at today's price (ignoring interest). By symmetry  $\mathbb{E}[\max(-Z, 0)]$  is the same, and  $|Z| = \max(Z, 0) + \max(-Z, 0)$ , so

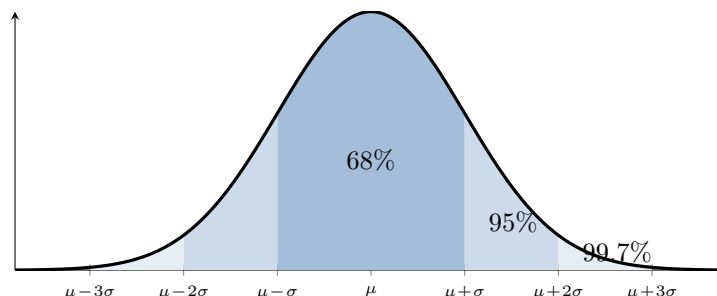
$$\mathbb{E}|Z| = \frac{2}{\sqrt{2\pi}} = \sqrt{\frac{2}{\pi}} \approx 0.80 :$$

a normal variable lands, on average, about  $0.8\sigma$  away from its mean.

## 14.4 The empirical rule

**Theorem 14.8.** If  $X \sim \text{Normal}(\mu, \sigma^2)$  then

$$\begin{aligned} \mathbb{P}\{\mu - \sigma \leq X \leq \mu + \sigma\} &\approx 68\%, \\ \mathbb{P}\{\mu - 2\sigma \leq X \leq \mu + 2\sigma\} &\approx 95\%, \\ \mathbb{P}\{\mu - 3\sigma \leq X \leq \mu + 3\sigma\} &\approx 99.7\%. \end{aligned}$$



**Remark 14.9.** This is worth internalizing because it turns  $\sigma$  into an interpretable number. If a process has  $\mu = 60$  and  $\sigma = 10$ , then roughly 95% of output lies in  $[40, 80]$ , and a value of 100 is a four-sigma event — about 1 in 30,000. The rule is also the fastest sanity check on an exam: if your answer says  $\mathbb{P}\{X > \mu + \sigma\} = 0.40$ , you have made an arithmetic error, because it must be about 0.16.

**Example 14.10** (How rare is a bad day?). A stock's daily return, in percent, is modeled as  $\text{Normal}(0, 1)$ , and a year has 252 trading days. How often should we see a one-day loss of more than 2%? Of more than 5%?

A loss of more than 2% is the event  $Z < -2$ , with probability  $\Phi(-2) = 0.0228$ ; over a year that is  $252 \times 0.0228 \approx 6$  days. A loss of more than 5% has probability  $\Phi(-5) \approx 2.9 \times 10^{-7}$  (off the end of the usual table): about one day in 3.5 million, or once in roughly 14,000 years of trading.

**Remark 14.11** (Fat tails). Stock markets have had many days with falls of more than 5% in the past forty years, and on October 19, 1987 the US market fell more than 20% in a single day. The normal model describes typical days well and extreme days badly: real returns have *fat tails*. The lesson is general. When a normal model says an event should happen once in ten thousand years and you have seen it several times, doubt the model, not your luck.

## 14.5 Looking ahead: the normal approximation to the binomial

If  $X \sim \text{Binomial}(n, p)$  then  $\mu = np$  and  $\sigma^2 = np(1 - p)$ . For large  $n$  the binomial pmf traces out a normal curve, and

$$\mathbb{P}\{X \leq x\} \approx \Phi\left(\frac{x + 0.5 - np}{\sqrt{np(1 - p)}}\right).$$

The  $+0.5$  is the **continuity correction**: we are approximating a lattice-valued variable by a continuous one, so the discrete point  $x$  is represented by the interval  $[x - 0.5, x + 0.5]$ . The usual rule of thumb is that the approximation is good when  $np \geq 10$  and  $n(1 - p) \geq 10$ .

We will see in Lecture 21 that this is just the central limit theorem applied to  $X = \mathbf{1}_1 + \cdots + \mathbf{1}_n$ , where  $\mathbf{1}_i$  is the indicator of a success on the  $i$ -th trial: a binomial count is a sum of  $n$  i.i.d. terms.